

LAB#13

POINTERS IN C++ PROGRAMMING

Lab Tasks

- 1- What would be the output of the following function, point out the errors, if any.

```
#include <iostream>
using namespace std;

int main ()
{
    int firstvalue = 5, secondvalue = 15;
    int * p1, * p2;

    p1 = &firstvalue; // p1 = address of firstvalue
    p2 = &secondvalue; // p2 = address of secondvalue
    *p1 = 10;          // value pointed to by p1 = 10
    *p2 = *p1;         // value pointed to by p2 = value pointed by p1
    p1 = p2;           // p1 = p2 (value of pointer is copied)
    *p1 = 20;          // value pointed by p1 = 20

    cout << "firstvalue is " << firstvalue << '\n';
    cout << "secondvalue is " << secondvalue << '\n';
    return 0;
}
```

- 2- Write a C++ program by using pointer that receives array of 5 integers from keyboard and calculate the sum, average and standard deviation of these numbers.
- 3- Modify the solution of task 2 in order to print the elements of the array in reverse order by using a pointer.
- 4- Write a program that asks the user to enter integers as inputs to be stored in the variables 'h' and 'j' respectively. There are also two integer pointers named ptrH and ptrJ. Assign the values of 'h' and 'j' to ptrH and ptrJ respectively, and display them.
- 5- Write a C++ program to find the max of an integral data set. The program will ask the user to input the number of data values in the set and each value. The program prints on screen a pointer that points to the max value.
- 6- Take input in variable and display same value by pointer.

- 7- Given the string "A string." Print on one line the letter on the index 0, the pointer position and the letter t. Update the pointer to pointer +2. Then, in another line print the pointer and the letters r and g of the string (using the pointer).

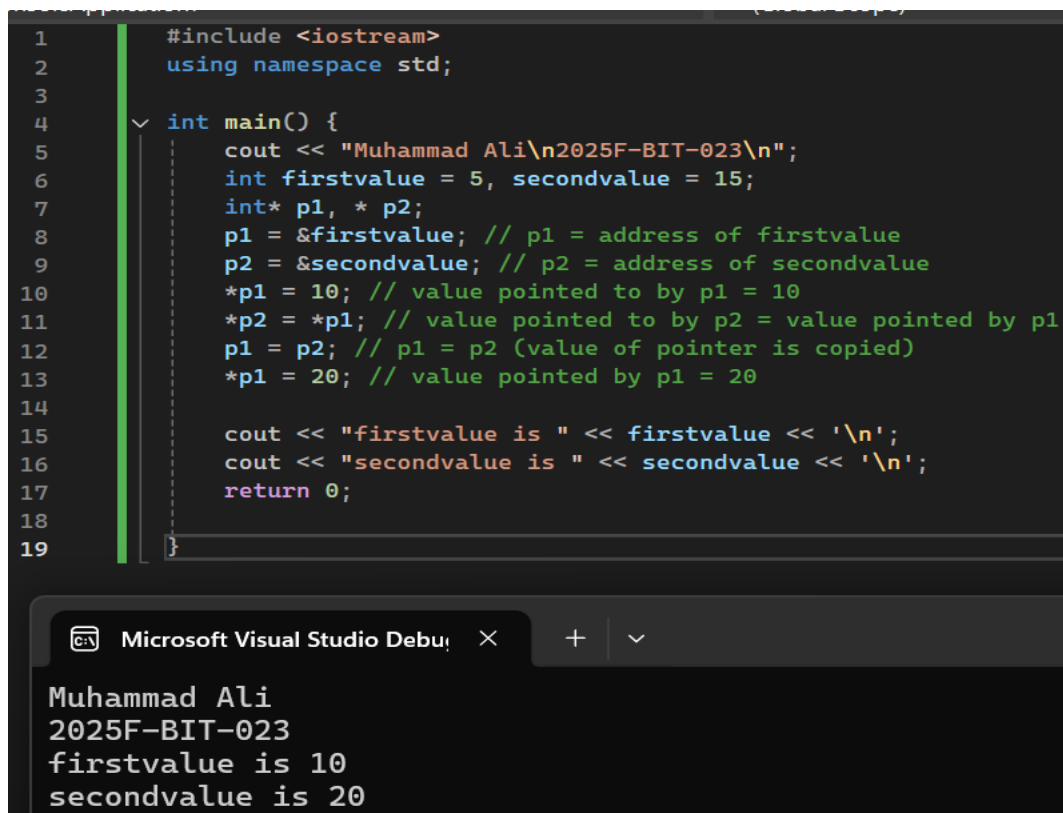
- 1- What would be the output of the following function, point out the errors, if any.

```
#include <iostream>
using namespace std;

int main ()
{
    int firstvalue = 5, secondvalue = 15;
    int * p1, * p2;

    p1 = &firstvalue; // p1 = address of firstvalue
    p2 = &secondvalue; // p2 = address of secondvalue
    *p1 = 10;          // value pointed to by p1 = 10
    *p2 = *p1;         // value pointed to by p2 = value pointed by p1
    p1 = p2;           // p1 = p2 (value of pointer is copied)
    *p1 = 20;          // value pointed by p1 = 20

    cout << "firstvalue is " << firstvalue << '\n';
    cout << "secondvalue is " << secondvalue << '\n';
    return 0;
}
```



The screenshot shows a C++ program in Visual Studio. The code defines two integer variables, `firstvalue` and `secondvalue`, and two pointers, `p1` and `p2`. `p1` is assigned the address of `firstvalue`, and `p2` is assigned the address of `secondvalue`. The value at `*p1` is set to 10, and then `*p2` is set to the value at `*p1` (10). Then, `p1` is assigned the value of `p2` (the address of `secondvalue`), and finally, `*p1` is set to 20. The program prints the values of `firstvalue` and `secondvalue`.

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      cout << "Muhammad Ali\n2025F-BIT-023\n";
6      int firstvalue = 5, secondvalue = 15;
7      int* p1, * p2;
8      p1 = &firstvalue; // p1 = address of firstvalue
9      p2 = &secondvalue; // p2 = address of secondvalue
10     *p1 = 10; // value pointed to by p1 = 10
11     *p2 = *p1; // value pointed to by p2 = value pointed by p1
12     p1 = p2; // p1 = p2 (value of pointer is copied)
13     *p1 = 20; // value pointed by p1 = 20
14
15     cout << "firstvalue is " << firstvalue << '\n';
16     cout << "secondvalue is " << secondvalue << '\n';
17     return 0;
18 }
19
```

The output window shows the following text:

```
Muhammad Ali
2025F-BIT-023
firstvalue is 10
secondvalue is 20
```

- 2- Write a C++ program by using pointer that receives array of 5 integers from keyboard and calculate the sum, average and standard deviation of these numbers.

```
#include <iostream>
using namespace std;
int main() {
    cout << "Muhammad Ali\n2025F-BIT-023\n";

    int arr[5];
    int* ptr = arr;
    int sum = 0;
    float ave = 0, stan = 0;
    cout << "Enter 5 integers:\n";
    for (int i = 0; i < 5; i++) {
        cin >> *(ptr + i);
        sum += *(ptr + i);
    }

    ave = sum / 5.0;
    float variance = 0;
    for (int i = 0; i < 5; i++) {
        variance += pow(*(ptr + i) - ave, 2);
    }

    variance /= 5;
    stan = sqrt(variance);
    cout << "Sum: " << sum << endl;
    cout << "Average: " << ave << endl;
    cout << "Standard Deviation: " << stan << endl;
    return 0;
}
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023
Enter 5 integers:
1
2
3
4
5
Sum = 15
Average = 3
Standard Deviation = 1.41421

- 3- Modify the solution of task 2 in order to print the elements of the array in reverse order by using a pointer

```

//
#include <iostream>
using namespace std;
int main() {
    cout << "Muhammad Ali\n2025F-BIT-023\n";

    int arr[5];
    int* ptr = arr;
    int sum = 0;
    float ave = 0, stan = 0;
    cout << "Enter 5 integers: \n";
    for (int i = 0; i < 5; i++) {
        cin >> *(ptr + i);
        sum += *(ptr + i);
    }
    ave = sum / 5.0;
    float variance = 0;
    for (int i = 0; i < 5; i++) {
        variance += pow(*(ptr + i) - ave, 2);
    }
    variance /= 5;
    stan = sqrt(variance);
    cout << "Sum: " << sum << endl;
    cout << "Average: " << ave << endl;
    cout << "Standard Deviation: " << stan << endl;
    cout << "Reverse order: ";
    for (int i = 4; i >= 0; i--) {
        cout << *(ptr + i) << " ";
    }
    return 0;
}

```

Microsoft Visual Studio Debug Console

```

Muhammad Ali
2025F-BIT-023
Enter 5 integers:
1
2
3
4
5
Sum: 15
Average: 3
Standard Deviation: 1.41421
Reverse order: 5 4 3 2 1

```

- 4- Write a program that asks the user to enter integers as inputs to be stored in the variables 'h' and 'j' respectively. There are also two integer pointers named ptrH and ptrJ. Assign the values of 'h' and 'j' to ptrH and ptrJ respectively, and display them.

```
#include <iostream>
using namespace std;

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023\n";
    int h, j;
    cout << "Enter the value of two integers: ";
    cin >> h >> j;
    int *prth, *prtj;
    prth = &h;
    prtj = &j;
    cout << "Value of h: " << *prth << endl;
    cout << "Value of j: " << *prtj << endl;
    return 0;
}
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023
Enter the value of two integers: 6
7
Value of h: 6
Value of j: 7

- 5- Write a C++ program to find the max of an integral data set. The program will ask the user to input the number of data values in the set and each value. The program prints on screen a pointer that points to the max value.

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      cout << "Muhammad Ali\n2025F-BIT-023\n";
6
7      int size;
8      cout << "Enter number of values: ";
9      cin >> size;
10     int data[100];
11     cout << "Enter the values:\n";
12     for (int i = 0; i < size; i++) {
13         cin >> data[i];
14     }
15     int* ptr = data;
16     int* ptrMax = &data[0];
17     for (int i = 1; i < size; i++) {
18         if (*(ptr + i) > *ptrMax) {
19             ptrMax = (ptr + i);
20         }
21     }
22     cout << "Max value: " << *ptrMax << endl;
23     cout << "Address of max value: " << ptrMax << endl;
24     return 0;
25 }
```

Microsoft Visual Studio Debug Console

```
Muhammad Ali
2025F-BIT-023
Enter number of values: 5
Enter the values:
1
2
3
4
5
Max value: 5
Address of max value: 000000DBCAEFF4B0
```

- 6- Take input in variable and display same value by pointer.

```
#include <iostream>
using namespace std;

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023\n";
    int num;
    cout << "Enter Number: ";
    cin >> num;
    int* ptr;
    ptr = &num;
    cout << "Value by pointer: " << *ptr;
    return 0;
}
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023
Enter Number: 4
Value by pointer: 4

- 7- Given the string "A string." Print on one line the letter on the index 0, the pointer position and the letter t. Update the pointer to pointer +2. Then, in another line print the pointer and the letters r and g of the string (using the pointer).

```
#include <iostream>
using namespace std;

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023\n";

    char str[] = "A string.";
    char* ptr = str;
    cout << "Index 0: " << str[0] << endl;
    cout << "Pointer position: " << ptr << endl;
    cout << "Letter t: " << *(ptr + 3) << endl;
    ptr = ptr + 2;
    cout << "Pointer +2: " << ptr << endl;
    cout << "Letter r: " << *(ptr + 2) << endl;
    cout << "Letter g: " << *(ptr + 5) << endl;
    return 0;
}
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023
Index 0: A
Pointer position: A string.
Letter t: t
Pointer +2: string.
Letter r: r
Letter g: g

